Reptile-Assisted Emotional and Behavioral Management in Individuals with Autism Spectrum Disorder

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Abstract

This study conducts a literature-based investigation into the biological foundations of emotional and behavioral disorders and examines the behavioral characteristics of reptiles as a potential modality within animal-assisted therapy. It further reviews contemporary societal attitudes toward reptiles from ethical, legal, and cultural perspectives. By integrating biological, therapeutic, and socio-cultural dimensions, this research establishes a theoretical foundation for the application of reptile-assisted intervention in therapeutic environments.

Chapter 1 - Introduction

A review of psychological and animal-assisted therapy (AAT) literature reveals that reptiles have demonstrated unique therapeutic potential in interventions targeting emotional and behavioral disorders. Compared to highly interactive mammals such as dogs—whose expressive and spontaneous behaviors can sometimes be overwhelming—reptiles are characterized by predictability, behavioral stability, and low intrusiveness. Their slow, deliberate movements and calm demeanor create a controlled and non-threatening environment, which is particularly beneficial for individuals with sensory sensitivities or those prone to overstimulation, such as individuals with autism.

Furthermore, as hairless animals, reptiles offer a hypoallergenic alternative for patients who are allergic to fur-bearing species, enabling safe tactile and emotional engagement. Studies suggest that gentle interactions with reptiles—through observation, handling, or caregiving—can foster patience, focus, and self-regulation, while supporting emotional awareness and behavioral control. Consequently, reptiles introduce a quieter, more predictable form of human–animal interaction within the broader field of animal-assisted therapy, expanding its accessibility and offering new therapeutic possibilities for individuals with autism and emotional dysregulation.

1.1 Background of the problem

In contemporary society, a growing number of individuals are affected by conditions commonly associated with modern lifestyles, including emotional regulation difficulties, behavioral disorders, and autism spectrum disorder. These conditions not only influence psychological well-being but also significantly affect social functioning, learning capacity, and long-term adaptability. Conventional treatment approaches primarily rely on human-centered interventions such as psychological counseling, cognitive guidance, and behavioral therapy.

In recent years, animal-assisted therapy (AAT) has emerged as an alternative and complementary intervention, demonstrating positive effects on emotional regulation, social engagement, and anxiety reduction. Existing AAT practices largely focus on mammals such as dogs, horses, and cats. However, these animals may not be suitable for all patients due to factors such as sensory sensitivity, fear of close physical contact, allergies to fur, or difficulties with unpredictable behavior—especially among individuals with autism spectrum disorder.

Consequently, there is an increasing need to explore alternative therapeutic agents that are less invasive, more predictable, and suitable for patients with heightened sensory sensitivity.

1.2 Research Questions

This study is guided by the following key questions:

1. Can reptiles function as effective mediators of emotional safety and stress reduction within animal-assisted therapy?

2. How do patients with autism spectrum disorder and emotional or behavioral disorders respond emotionally and behaviorally to reptile-assisted interaction?

3. What therapeutic advantages and limitations distinguish reptiles from traditional mammalian therapy animals?

4. How can reptile-assisted theintegrated irapy be nto existing therapeutic environments in a safe, ethical, and structured manner?

5. How can keeping reptiles contribute to human well-being, and what ethical, legal, and ecological challenges does it raise?

1.3 ESP question

How can a therapeutic psychological space be designed to support patients with emotional and behavioral dysregulation in better engaging with treatment?

1.4 Research Objective

This study aims to explore the therapeutic potential of reptiles as an alternative form of animal-assisted intervention. By focusing on the non-invasive, gentle, predictable, and hairless characteristics of reptiles, this research seeks to examine their feasibility for integration into animal-assisted therapy frameworks. The ultimate objective is to contribute to the development of more diversified, inclusive, and adaptive therapeutic environments that address the specific needs of individuals with emotional, behavioral, and neurodevelopmental disorders.

1.5 Design Positioning

This research adopts a mixed-methods approach combining qualitative and quantitative strategies to investigate the potential of reptile-assisted intervention as a form of animal-assisted therapy. The study is primarily grounded in an extensive literature review covering psychology, neuroscience, animal-assisted therapy, ethology, and socio-cultural studies in order to establish an interdisciplinary theoretical foundation. In parallel, a questionnaire survey was conducted to examine public perceptions, emotional responses, risk awareness, and acceptance of reptiles and reptile-assisted therapy within the Nantes region. Semi-structured in-depth interviews were further carried out with psychotherapists, special education professionals, social workers, reptile keepers, and parents of individuals with emotional, behavioral, or developmental disorders to obtain professional, experiential, and ethical perspectives. In addition, self-testing and participatory observation were employed as embodied research methods, allowing the researcher to directly experience and record emotional, sensory, and psychological responses during controlled interactions with reptiles in simulated therapeutic settings. Field observation in therapeutic, educational, and public animal-related environments was also conducted to analyze interaction patterns, environmental conditions, and safety factors. Quantitative data were analyzed using descriptive statistical methods, while qualitative data were examined through thematic analysis. Ethical principles including informed consent, participant anonymity, and non-invasive animal welfare protocols were strictly followed throughout the research process.

1.6 Research Scope (Nantes Region)

This research is geographically situated in the Nantes metropolitan area and its surrounding regions. The study is conducted through local field observations, interviews with psychotherapists and mental health professionals in Nantes, and qualitative observation of patients with autism spectrum disorder as well as individuals experiencing emotional and behavioral regulation difficulties. By grounding the research in a localized healthcare and social context, the study aims to ensure both contextual relevance and practical applicability.

1.7 Overall exploration of each chapter

1.7.1 Chapter 2

Chapter 2 provides a comprehensive review of the existing literature on the neurobiological foundations of emotion and behavior, with a particular focus on the regulatory role of the nervous system in emotional processing and behavioral control. It first examines how neural mechanisms—especially the autonomic nervous system and right-hemispheric brain functions—shape emotional arousal, affective communication, and impulse regulation. The chapter further explores how parental behaviors and early caregiving patterns during infancy critically influence emotional development, attachment formation, and long-term emotional and behavioral outcomes across the lifespan.

In addition, this chapter systematically reviews the theoretical and empirical foundations of Animal-Assisted Therapy (AAT), highlighting its well-documented positive effects on emotional regulation, behavioral stabilization, stress reduction, and social functioning. Special attention is given to the emerging role of reptiles as alternative therapeutic animals within AAT frameworks, emphasizing their unique advantages in predictability, low sensory stimulation, and non-invasive interaction, particularly for individuals with heightened sensory sensitivity.

The chapter also clarifies the clinical definition of autism spectrum disorder (ASD) and reviews key findings regarding its core characteristics, including deficits in social communication, emotional regulation difficulties, sensory processing atypicalities, and rigid behavioral patterns. By integrating neuroscientific, developmental, clinical, and therapeutic perspectives, this chapter establishes a multidisciplinary theoretical foundation for understanding how emotional and behavioral regulation develops, how it can be disrupted, and how it may be restored through animal-assisted and reptile-assisted interventions.

1.7.2 Chapter 3

Chapter 3 presents the methodological framework adopted in this study, integrating both quantitative and qualitative research approaches to investigate the potential of reptile-assisted therapy and public perceptions of reptiles as therapeutic animals. A mixed-method design was employed in order to capture both measurable tendencies and in-depth experiential insights.

The quantitative component consists primarily of structured questionnaire surveys aimed at collecting data on public attitudes toward reptiles, levels of acceptance, perceived risks, emotional responses, and ethical concerns related to reptile keeping and reptile-assisted therapy. This approach allows for the identification of general trends, behavioral patterns, and societal perceptions across a broader population.

The qualitative component is based on semi-structured in-depth interviews conducted with reptile keepers, breeders, therapists, and individuals with direct experience in reptile care or animal-assisted practices. These interviews explore personal motivations for keeping reptiles, emotional bonds with non-traditional companion animals, ethical positions, legal awareness, and the perceived therapeutic value of reptiles. Additional narrative interviews with members of the general public further examine cultural representations of reptiles, fear, stigma, curiosity, and evolving social attitudes toward exotic animals.

By combining quantitative attitudinal data with qualitative experiential narratives, this chapter aims to construct a multi-layered understanding of how reptiles are socially perceived, emotionally experienced, and ethically positioned within contemporary society. This methodological triangulation strengthens the validity of the research and provides a robust empirical foundation for the subsequent design exploration of reptile-assisted therapeutic spaces.

Chapter 2 - Neuro-Emotional and Behavioral Regulation in Animal-Assisted Therapy

2.1 Literature about Animal-Assisted Therapy

Animal-Assisted Therapy (A-AT), as a non-invasive adjunctive intervention grounded in attachment relationships and emotional co-regulation mechanisms, has demonstrated stable and significant therapeutic effects across psychology, medicine, education, and social work. A substantial body of research indicates that therapy animals—particularly dogs—can effectively reduce anxiety, stress, and pain levels within short periods, while simultaneously enhancing emotional stability, behavioral regulation, and social interaction (Dawn A. Marcus, 2013; Tracy S. Geist, 2011). These effects are evident not only in subjective psychological well-being but also in objective physiological indicators, including heart rate, blood pressure, cortisol, endorphin, and oxytocin levels (Marcus, 2013).

At the neuropsychological level, attachment theory provides the central explanatory framework for A-AT. As originally proposed by John Bowlby, secure attachment shapes emotional regulation through stable co-regulatory mechanisms, whereas insecure attachment undermines individuals' ability to integrate stress, relational experience, and self-worth. Building on affective neuroscience, Allan Schore demonstrated that emotional regulation is primarily organized in the right hemisphere and dependent on early attachment-mediated co-regulation (Schore, 2001; 2003; 2009). Through stable, non-judgmental, and consistently responsive interaction, therapy animals function as alternative attachment figures, reactivating right-hemispheric emotional regulation systems and thereby promoting emotional stabilization, self-integration, and behavioral self-control. From a cognitive perspective, Aaron T. Beck's schema theory further suggests that interaction with animals can interrupt automatic negative cognitive chains and repair dysfunctional self–other–environment schemas (Beck & Clark, 2011).

At the educational and behavioral levels, the "Companion Zoo (CZ)" program implemented by Aaron M. Katcher and William Wilkins at the Devereux Center provides strong empirical evidence that embodied, empathy-oriented animal interaction environments significantly enhance classroom attendance, learning efficiency, behavioral control, and perceived environmental mastery among students with special needs, while markedly reducing aggression and emotional dysregulation (Katcher & Wilkins, 1998). Although some behavioral improvements diminish after students leave the animal-assisted environment, the findings clearly indicate that the animal-centered context itself plays a decisive role in behavioral transformation.

In the domains of trauma recovery and emotional regulation, therapy dogs have been successfully applied to adolescents with histories of family violence, abuse, aggressive behavior, and emotional disorders. Through emotional bonding with animals, narrative reflection, and guided interaction, adolescents show significant increases in safety, self-worth, well-being, and empathy, alongside marked reductions in loneliness and stress (Geist, 2011). In these processes, animals function not merely as companions but as mediators of personality repair and emotional–cognitive restructuring.

The Hill Top Academy case further illustrates that therapy dogs can operate as non-verbal emotional regulators during crisis states. Through bodily proximity, behavioral feedback, and emotional mirroring, dogs effectively replace physical restraint, help students transition from emotional collapse to self-regulation, and facilitate the transfer of emotional stability into therapeutic relationships (Geist, 2011). Trauma research indicates that traumatic memories are primarily stored in the right hemisphere and are difficult to process verbally (Schore, 2003; Siegel, 1999). Therapy animals communicate directly through non-verbal channels aligned with right-hemispheric processing, thereby offering safe attachment experiences that gradually generalize into human relationships.

From a physiological perspective, dogs' sensitivity to disease and emotional states is largely attributed to their superior olfactory system. Stress and illness induce biochemical changes—including cortisol, catecholamines, cytokines, and growth factors—which are transmitted through body odors and can be detected by dogs (Marcus, 2013). At the same time, studies show that therapy dogs also experience positive emotional feedback during therapeutic interactions, as reflected by increased endorphins and dopamine, while cortisol levels remain only mildly elevated, indicating that A-AT is ethically sustainable from an animal welfare perspective (Marcus, 2013).

From an evolutionary and socio-historical standpoint, emotional relationships between humans and animals are not modern inventions. Archaeological evidence demonstrates that affective human–animal bonds existed as early as the hunter-gatherer period, including a 12,000-year-old burial in Israel in which a human was interred with a puppy in an explicitly affectionate posture (Davis & Valla, 1978). This deep evolutionary continuity suggests that animal-mediated emotional regulation and social bonding constitute a long-standing adaptive mechanism rather than a transient cultural phenomenon.

Crucially, beyond dogs, **reptile-assisted support has also demonstrated measurable therapeutic effects**. A controlled behavioral study by Francie R. Murry and M. Todd Allen confirmed that reptile-assisted support significantly reduces both internalizing and externalizing problem behaviors in female children with emotional disturbance (Murry & Allen, 2012). This finding provides the first empirical validation that **low-arousal, non-mammalian species can also serve as effective emotional regulators within therapeutic contexts**.

Taken together, existing research confirms that animal-assisted therapy possesses solid scientific foundations across multiple dimensions, including emotional recovery, behavioral regulation, attachment reconstruction, cognitive restructuring, and social adaptation. Its core value extends far beyond companionship, representing a systematic intervention method through which cross-species relationships help reorganize emotional order, self-concept, and social connectedness. This makes A-AT particularly suitable for individuals with insecure attachment, traumatic backgrounds, attention deficits, emotional disorders, and behavioral dysregulation. At the same time, this extensive theoretical and empirical foundation provides a strong translational basis for the future integration of **low-invasiveness, non-verbal, and low-arousal species such as reptiles** into animal-assisted therapeutic frameworks.

2.2 Emotion and Behavior as a Bio–Relational–Cognitive Regulation System

In my view, human emotion and behavior are not determined solely by personality traits or willpower, but emerge from a dynamic regulatory system jointly shaped by neurobiological mechanisms, attachment structures, and cognitive processing.

The right hemisphere of the brain and the autonomic nervous system are primarily responsible for emotional arousal, empathy, and non-verbal communication, while the cognitive system influences emotional and behavioral outputs through beliefs, schemas, and meaning-making processes. When this regulatory system is disrupted during early attachment development, or distorted later through maladaptive cognitive processing, individuals may exhibit emotional dysregulation, impulsive behavior, anxiety, autism-related symptoms, or social withdrawal.

Therefore, emotional and behavioral disorders should not be understood as mere “bad habits,” but as the result of a dysbalanced regulatory system.

2.3 Attachment as a Cross-Species System of Emotional Regulation

I adopt the core transformation of attachment theory that emerged in the 1970s: attachment is neither purely instinctual nor solely learned, but an integrated regulatory system combining biological predispositions, social learning, perception, motivation, and emotion. This system is not exclusive to humans; it exists in homologous forms across many species.

This implies that:

The origin of emotional stability lies not simply in human relationships themselves, but in whether a predictable, responsive, and secure regulatory object is present.

Thus, the attachment system is inherently capable of cross-species transfer, which constitutes the fundamental biological legitimacy of animal-assisted therapy.

2.4 Emotional Disorders and Autism as Failures of Regulation Rather Than of Emotion

From the perspective of right-hemispheric emotional regulation and the Generic Cognitive Model (GCM), I argue that:

- Emotional disorders do not result from "too much emotion," but from the failure to integrate and regulate emotion.
- Autism does not imply emotional absence, but rather atypical functioning in non-verbal emotional communication, empathy, and predictive social processing.
- Behavioral dysregulation does not reflect weak willpower, but rather the dominance of automatic reactive systems over reflective regulatory systems.

In essence, the core problem is not whether emotion exists, but whether individuals possess a safe, stable, and low-stress emotional regulation pathway.

2.5 The Core of Animal-Assisted Therapy Is Emotional Co-Regulation, Not Mere Companionship

Within my framework, Animal-Assisted Therapy (AAT) is not merely a form of emotional comfort or companionship, but a biological–neural–emotional co-regulation mechanism in its own right.

Through:

- Slow and rhythmic physiological patterns
- Stable behavioral feedback
- Non-verbal interaction
- High predictability

AAT directly modulates:

- The vagus nerve system
- Emotional arousal thresholds
- Neurobiological safety circuits
- Self-regulatory capacity

Its essential function is:

To reactivate emotional regulation through interaction with a living being that does not judge, demand, or impose complex social pressure.

2.6 Why Reptiles: Minimal Stimulation, Maximum Predictability, and Low Invasiveness

Among all therapy animals, I argue that reptiles possess unique regulatory advantages for individuals with autism and emotional or behavioral disorders:

- They lack complex facial expressions, reducing social anxiety.
- Their slow and rhythmic movements prevent neural overstimulation.
- They do not actively demand emotional interaction, thus minimizing defensive responses.
- Their behavior is highly predictable, helping to reconstruct the belief that "the world is safe."

These characteristics directly address the core needs of individuals with autism and emotional dysregulation:

Low stimulation, low invasiveness, stability, safety, and controllability.

Thus, reptiles are not emotionally "cold substitutes," but rather neurobiologically gentle and highly compatible regulatory mediators.

2.7 Autism

Autism is characterized by difficulties in recognizing and managing social cues, as well as challenges related to sensory processing. Individuals with autism often react abnormally to sensory stimuli, being either hypersensitive or hyposensitive. Studies show that 75% of autistic children have sensory issues.
Most individuals with autism are either hypersensitive or hyposensitive to light, sound, crowds, and other external stimuli. Some may display both types of sensitivity at once. This often leads autistic individuals to cover their ears, avoid brightly lit environments, or, conversely, crash into sofas and crave strong physical contact such as tight hugs.

Tactile hypersensitivity is also common. Individuals who are hypersensitive to touch often show strong resistance to social contact and avoid being kissed, hugged, or touched. They may be unable to wear certain types of clothing and struggle with certain textures. Some foods can be hard for them to tolerate, and they may feel discomfort using high-sensory-output items like combs, brushes, and socks.

2.8 Reptiles and Autism

Reptiles' calm and quiet nature gives them a soothing and therapeutic effect on children with autism. Their gentle temperament, predictable behavior, and manageable size make them suitable companions for children who are easily overwhelmed by more energetic or unpredictable pets.

Bearded dragons, for example, thrive on routine, and their serene presence provides a sense of stability and predictability for children with autism or sensory issues. This kind of companionship can help reduce anxiety, offer emotional support, and assist children in coping with the challenges of autism.

Children with Autism Spectrum Disorder (ASD) and somatosensory processing disorders tend to avoid furry animals, such as dogs, cats, rabbits, and guinea pigs, but are more likely to form deeper connections with reptiles, whose skin is either smooth or has discernible textures.

2.9 Support on the Internalizing and Externalizing Behaviors of Female Children with Emotional Disturbance

A study investigated children who had lost one or both parents, using reptiles in animal-assisted therapy to provide emotional support, train caregiving skills, and help them learn emotional expression.

Research shows that children who participated in reptile-assisted therapy demonstrated improvements in multiple areas, including:

- withdrawal behaviors,
- somatic (physical) symptoms,
- aggressive and rule-breaking behaviors,
- as well as enhanced social skills, attention, and cognitive abilities.

After structured interactions with animals, individuals with mental health conditions experienced fewer episodes of aggression, anxiety, and mood disturbances.

Animals are also often used as comforting companions for bereaved children. In the United States, by the age of 15, at least 5% of children (approximately 1.5 million) have experienced the death of a parent (Silverman & Worden, 1992).

Losing a parent makes children more vulnerable to trauma and psychological distress, with effects far greater than those observed among their peers who have not experienced such loss. This grief-related adversity may manifest as:

- significant depression, anxiety, aggression, or other disruptive behaviors,

- poor academic performance, impaired social functioning, and low self-concept.

Compared to boys, girls are more likely to develop conduct problems after parental loss.
Other common outcomes include difficulty forming close relationships, learning difficulties, depression, and somatic complaints such as physical discomfort.

According to the U.S. federal definition of emotional disturbance in educational settings, the condition is characterized by one or more of the following:
1. An inability to learn that cannot be explained by intellectual, sensory, or health factors.
2. An inability to build or maintain satisfactory interpersonal relationships with peers and teachers.
3. Inappropriate types of behavior or feelings under normal circumstances.
4. A general pervasive mood of unhappiness or depression.
5. A tendency to develop physical symptoms or fears associated with personal or school problems.

Mechanisms of behavioral improvement
1. Caring for animals — feeding, cleaning, interacting, and identifying emotional cues.
2. Learning non-violent communication, improving attention and empathy, and developing consistent routines.
3. Generalizing this caregiving behavior toward animals into other social environments.

2.10 Physical improvements

Primary caregivers reported that children experienced fewer headaches, better eating habits, and reduced vomiting after participating in reptile-assisted sessions.

2.11 Why were anxiety and depression not significantly improved?

Researchers suggest that the combination of parental loss and frequent transitions in educational or living environments represents a major trauma that cannot be fully alleviated by 16 weeks of animal-assisted support alone.
These symptoms require longer-term and more complex psychological intervention.

In a non-judgmental group environment, girls were able to express their emotions more openly and establish healthier behavioral habits.

Chapter 3 - Practical Case Studies of Animal-Assisted Therapy

3.1 Bearded dragons as therapeutic animals - A case study of autism and a bearded dragon

Throughout human evolutionary history, relationships between humans and animals have been deeply embedded in survival and social structures. Archaeological evidence, including prehistoric cave paintings, frequently depicts scenes of humans gathered with wolves around campfires and collaborating with dogs and horses during hunting activities. These early visual records suggest that interspecies cooperation played a critical role in enhancing survival efficiency and environmental adaptation for both humans and animals. As Serpell (1996) noted, *"the domestication of animals was not merely a technical innovation, but a profound social and emotional transformation in human history"* (p. 21).

With the development of civilization, animals gradually assumed diversified functional roles within human societies, including hunting assistance, agricultural labor, protection, companionship, and entertainment. Importantly, human attitudes toward animals are not uniform but are shaped by species-specific characteristics and culturally constructed beliefs. Some animals are perceived as productive partners, others as companions, and still others as objects of fear or symbolic meaning.

The therapeutic potential of animals was formally recognized as early as the nineteenth century. Florence Nightingale, through systematic observation in psychiatric institutions, reported that the presence of small animals helped reduce agitation and anxiety among patients. This marked one of the earliest empirical acknowledgments of what would later be conceptualized as Animal-Assisted Therapy (AAT).

Contemporary animal-assisted practices have further demonstrated structured therapeutic frameworks. In Israel, for instance, therapists at organizations such as Ahavat Zion implement animal-assisted programs for children aged 6 to 12 presenting with communication disorders, social difficulties, aggressive behavior, and autism spectrum conditions. Children participating in these programs are introduced to animals within carefully regulated environments governed by strict behavioral rules, including prohibitions on shouting, running, and violating animals' personal space. These rules establish mutual respect and clear behavioral boundaries, which are essential for emotional regulation and therapeutic safety.

Today, AAT encompasses a wide range of animal species. While small animals such as rabbits, birds, fish, and rodents are widely used due to their low physical risk and accessibility, larger animals—including horses, dolphins, elephants, and increasingly reptiles—are also incorporated depending on therapeutic objectives. This diversity reflects the growing understanding that different species activate distinct sensory, emotional, and cognitive regulatory pathways.

Katcher (1983) provided one of the earliest structural models for understanding the human–animal bond, identifying four foundational dimensions:

(1) Security, referring to the emotional safety derived from the animal's presence;

(2) Intimacy, mediated through physical contact such as touch and caressing;

(3) Proximity, in which the animal is treated as a family member; and

(4) Permanence, denoting the stable, enduring nature of the relationship.

Together, these historical, cultural, and psychological perspectives demonstrate that human–animal relationships are not peripheral phenomena, but long-standing regulatory systems that shape emotional security, behavioral stability, and social attachment across both evolutionary and therapeutic contexts.

Chapter 4 - An Investigation into Public Perceptions of Reptiles

4.1 Survey

A total of 33 questionnaire responses were collected from participants primarily aged between 18 and 30, with a small number of respondents under 18 and above 30. The geographic distribution was mainly concentrated in France (including Nantes, Pays de la Loire, Loire-Atlantique, and Mayenne), with additional responses from China, Australia, Mexico, and New Caledonia, reflecting a limited but international sample.

4.1.1 Reptile Ownership and Exposure

The large majority of respondents reported that they do not currently own reptiles. Most participants also stated that 0 to 2 people in their immediate social environment keep reptiles, indicating that reptile ownership remains a relatively marginal practice within their social circles. The primary reasons for not owning reptiles included:

- Housing and logistical constraints
- Financial limitations
- Fear or discomfort with reptiles
- Family disapproval
- Lack of time or stable living conditions

Only a very small number of respondents reported actual experience keeping reptiles (e.g., owning lizards).

4.1.2 Public Attitudes Toward Reptiles

Participants’ attitudes toward reptiles can be grouped into three dominant categories:

1. Neutral to Curious Attitudes
A significant portion of respondents expressed neutral curiosity, describing reptiles as:

- “Interesting”
- “Original”
- “Different from cats and dogs”

- "Atypical"

These respondents did not reject reptiles but viewed them as unconventional companion animals.

2. Positive and Aesthetic-Oriented Attitudes
Some respondents highlighted:

- The beauty of reptiles
- Their uniqueness
- Their low time-demand compared to dogs
- Their observational value rather than interactive companionship

These respondents associated reptiles with calmness, observation, and visual pleasure, rather than emotional bonding.

3. Negative or Fear-Based Attitudes
A smaller but notable group expressed:

- Fear ("creepy," "scary")
- Disgust
- The belief that reptiles are not "affectionate"
- The idea that reptiles should remain wild

These views reveal persistent cultural stereotypes and emotional distance toward reptiles.

4.1.3 Perceived Motivations for Keeping Reptiles

When asked why people keep reptiles, respondents most frequently mentioned:

- Originality and uniqueness ("It's original," "different from other pets")
- Personal passion
- Aesthetic appreciation
- Low maintenance needs
- Observation rather than emotional interaction
- Exotic appeal

Some responses also associated reptile keeping with:

- A desire to stand out
- Personal identity expression
- Atypical lifestyle preferences

4.1.4 Ethical and Cultural Perceptions

Several respondents explicitly mentioned ethical concerns, including:
- Discomfort with the idea of keeping wild animals in captivity
- The belief that reptiles "should stay wild"
- Uncertainty about animal welfare conditions

These responses indicate that ethical hesitation remains a significant barrier to the social acceptance of reptiles as companion animals.

4.1.5 Key Implications for Reptile-Assisted Therapy Research

Overall, the survey reveals that reptiles are currently perceived as:
- Socially marginal as companion animals
- Aesthetically interesting but emotionally distant
- Low-interaction and low-demand animals

At the same time, the strong association with calmness, low intrusion, and observation aligns precisely with the theoretical foundations of low-arousal, non-verbal therapeutic intervention. The presence of fear and ethical concern further highlights the importance of:
- Careful design of therapeutic environments
- Strong educational mediation
- Clear ethical and welfare frameworks

These findings suggest that reptile-assisted therapy faces both cultural resistance and strong design opportunities: while public familiarity and emotional acceptance remain limited, the perceived non-invasiveness and low-intensity interaction of reptiles directly support their potential therapeutic application for individuals with sensory sensitivity, emotional dysregulation, and autism spectrum disorder.

Keywords：
Animal-Assisted Therapy (AAT)
Individuals with Autism Spectrum Disorder (ASD)
Individuals with Emotional Regulation Difficulties
Individuals with Behavioral Regulation Difficulties

4.2 Interviews

4.2.1 Mickaël Grévois

a multifaceted perspective on reptile keeping that interweaves therapeutic practice, legal regulation, animal welfare ethics, and human–animal relational philosophy. Drawing on the experience of a licensed reptile keeper and additional reflective testimonies, the narrative reveals a clear distinction between responsible, welfare-oriented keepers and trend-driven, uninformed owners, emphasizing that reptile keeping requires long-term commitment, technical knowledge, and strict environmental control. While reptiles are shown to possess potential therapeutic value—particularly in supporting emotional regulation in children with behavioral difficulties through calm, non-verbal interaction—the text simultaneously highlights the profound ethical tensions surrounding captivity, commodification, selective breeding, and ecosystem disruption. Strict French and European legal frameworks are presented as necessary safeguards against ecological and welfare risks, especially concerning invasive and protected species. At a broader level, the discussion critically questions whether contemporary pet keeping reflects genuine care or human desires for control, novelty, and emotional substitution, drawing parallels between animal confinement and human experiences of isolation. Ultimately, the text argues that human–animal coexistence must be grounded in deep respect for animal needs, natural behavior, and environmental integrity, positioning animal care not merely as a technical practice, but as a fundamentally ethical and cultural responsibility.

4.2.2 Ayna

a childhood experience of keeping aquatic turtles as family pets and reveals how early human–animal interaction functions as an informal education in responsibility, empathy, and care. Initially introduced as a means for teaching children curiosity and accountability, the turtles gradually shifted from being perceived as simple pets to becoming full members of the family. Through daily maintenance, behavioral observation, health monitoring, and emotional attachment—particularly through the role assumed by the father—the experience demonstrated that turtles require complex environmental conditions, including clean water, temperature regulation, sunlight, adequate space, and careful feeding. The narrative also highlights risks associated with free roaming, spatial constraints, and impulsive pet acquisition, as well as the tendency to replace animals out of habit rather than reflective choice. Beyond individual experience, the account critically addresses broader social issues such as the commodification of pets, impulse-driven ownership, the pet industry’s aesthetic hierarchy, and the misconception that reptiles require minimal care. Ultimately, the case emphasizes that animals are not decorative objects but sentient beings with emotional and physical needs, and that ethical pet keeping should be grounded in long-term responsibility, respect, and genuine companionship rather than control or convenience.

4.2.3 Students from EDNA

This account highlights how contemporary pet-keeping is often driven less by deliberate ethical reflection and more by habit, social inertia, and symbolic attachment. Many individuals grow up with animals and continue keeping pets as a normalized life practice, frequently without establishing clear boundaries or long-term responsibility, particularly in family contexts involving children. At the same time, pets increasingly become objects of display and aesthetic consumption, shaped by an industry that constructs hierarchies of “premium” and “ordinary” animals based on appearance, rarity, and market value. However, fast-paced urban lifestyles, limited living space, and restrictive regulations often make it difficult to meet animals’ real needs, leading to neglect, abandonment, and frustration for both humans and animals. Under such conditions, pet ownership becomes highly constrained and regulated, with animals’ freedom increasingly shaped by governmental control rather than welfare-centered practice. The narrative ultimately suggests that modern pet-keeping has become, to some extent, a symbolic social behavior rather than a genuine response to animal needs, and that the well-being of companion animals today is largely defined by how human societies choose to regulate, manage, and spatially accommodate them.

4.2.4 Pet shop retailer

a complex and often contradictory contemporary landscape of reptile keeping shaped by urban living conditions, legal regulation, ethical debate, and personal passion. Reptiles are frequently chosen in cities because they are perceived as quiet, space-efficient, and compatible with restrictive public policies on pets, yet this convenience-driven logic often overlooks their emotional, environmental, and physiological needs. The interview reveals deep tensions between rescue intentions and ecological interference, between responsibility and impulsive ownership, and between conservation through captivity and the loss of natural freedom. While some keepers demonstrate lifelong commitment, professional knowledge, and legally regulated care, others acquire large or exotic species for social display or vanity, leading to neglect and abandonment as the animals grow. The narrative also highlights France’s particularly strict legal framework as a societal mechanism to control irresponsible behavior, contrasting it with more permissive systems elsewhere in Europe. Ultimately, the interview frames captivity as both a necessary tool for conservation, education, and biomedical survival (e.g., antivenom production), and a profound ethical dilemma, emphasizing that human–animal relationships must be grounded in responsibility, restraint, ecological awareness, and respect for animals’ natural behaviors rather than convenience, dominance, or symbolic possession.

Chapter 5 - Conclusion and Discussion

5.1 Discussion: Design Implications for Animal-Assisted Therapy Environments

The existing body of literature on Animal-Assisted Therapy (AAT) demonstrates that therapeutic efficacy does not arise solely from the presence of animals, but from a complex interaction between attachment mechanisms, neuropsychological regulation, cognitive restructuring, and environmental affordances. While current research has extensively validated the emotional, behavioral, and physiological benefits of AAT, far less attention has been given to the designed environment in which these therapeutic interactions occur. From a design perspective, the therapeutic outcome of AAT should be understood not only as a human–animal interaction, but as a human–animal–space system.

5.1.1 Spatial Design as a Regulator of Emotional Arousal

Neuroscientific and attachment-based studies (Schore; Siegel; Bowlby) emphasize that emotional regulation is primarily mediated by the right hemisphere and the autonomic nervous system. These systems are highly sensitive to sensory input, including light, sound, spatial density, temperature, visual complexity, and movement rhythms. Therefore, therapeutic space itself becomes an active co-regulatory agent.

Design can intentionally modulate:

- Arousal levels through lighting gradients, acoustic buffering, and visual softness
- Safety perception through enclosure, boundary clarity, and predictable spatial sequencing
- Attachment security through spatial consistency, recognizable routines, and stable object placement

From this perspective, the architectural and interior design of AAT clinics is not merely a background container but functions as a secondary attachment system that supports emotional stabilization alongside the animal.

5.1.2 From Cognitive Distortion to Spatial Cognition Repair

According to the Generic Cognitive Model (Beck & Clark), emotional disorders arise from biased information processing and maladaptive schemas. Therapy works by reconstructing perception–appraisal–emotion–behavior loops. In AAT environments, spatial interaction becomes an embodied cognitive intervention:

- Predictable layouts counter beliefs of environmental threat
- Slow, legible circulation paths reduce automatic hypervigilance
- Tangible interaction zones (care, feeding, observation) reinforce self-efficacy and agency

Thus, spatial design directly participates in schema reconstruction, supporting the cognitive dimension of emotional regulation described in CBT and GCM frameworks.

5.1.3 Design as a Mediator of Attachment Reconstruction

Attachment theory conceptualizes emotional healing as a process of repeated secure-base experiences. In AAT, animals function as alternative attachment figures. However, attachment is always situated in space. The environment determines:

- Whether interaction feels voluntary or forced
- Whether distance and proximity can be self-regulated
- Whether sensory safety is maintained

Design therefore mediates the quality of attachment experiences. For individuals with autism, trauma, or emotional disorders—who often exhibit hypersensitivity to noise, touch, and unpredictability—low-arousal, non-invasive spatial conditions become as essential as the animal itself.

5.1.4 Reptile-Assisted Therapy and Low-Arousal Design Logic

Compared with dogs or horses, reptiles introduce a fundamentally different therapeutic modality:

- Slow movement
- Minimal facial expression
- No fur, low odor, low vocalization
- High predictability

These qualities align with a low-arousal therapeutic design philosophy, making reptiles particularly compatible with:

- Autism spectrum disorder

- Sensory integration disorders
- Anxiety and emotional dysregulation

This implies specific spatial design requirements:
- Stable thermal microclimates
- Soft visual contrast
- Slow transitional zones between functions
- Transparent but protected enclosures enabling observation without intrusion

Reptile-assisted therapy therefore demands a new spatial typology, distinct from conventional dog-centered AAT clinics.

5.1.5 Educational and Behavioral Environments as Designed Systems

Case studies such as the Companion Zoo (Katcher & Wilkins) and the Hill Top Academy demonstrate that behavioral transformation is strongly context-dependent. The animal-centered environment itself functions as a behavioral modifier:
- Structuring routines
- Reinforcing responsibility
- Supporting emotional containment
- Enabling social mirroring

Yet when students exit the animal environment, improvements often attenuate—indicating that therapeutic effects are partly spatially embedded. This highlights the importance of designing repeatable, transferable spatial models for schools, clinics, and community spaces.

5.1.6 Toward a New Design Paradigm for Psychological Clinics

Taken together, the literature suggests that future psychological clinics integrating AAT should shift from:
- Diagnosis-centered rooms → co-regulation environments
- Therapist-dominated spaces → triadic human–animal–space systems
- Static consultation rooms → dynamic multisensory therapeutic landscapes

Design therefore becomes:
- A non-verbal therapist
- A regulator of autonomic arousal
- A scaffold for emotional learning
- A mediator between cognition, attachment, and behavior

5.1.7 Implications for Community-Based Therapeutic Architecture

In community contexts such as Nantes, the integration of reptile-assisted therapy into public or semi-public mental health spaces opens new design possibilities:

- Neighborhood-based therapeutic micro-centers
- Hybrid spaces combining care, education, and observation
- Gradual exposure environments reducing social anxiety
- Low-stigma mental health infrastructures embedded in daily life

Such environments could support children with autism, adolescents with behavioral disorders, and adults with emotional dysregulation through continuous, non-intrusive therapeutic contact, rather than isolated clinical interventions.

5.2 Conclusion

Building upon the integrated framework of attachment theory, cognitive psychopathology, animal-assisted intervention, and socio-ethical awareness, this research further proposes the design of a **safe, low-arousal, public therapeutic space within the Nantes community** to support individuals with emotional and behavioral disorders, as well as those on the autism spectrum. Rather than positioning therapy as an exclusively clinical or institutional process, the proposed space translates therapeutic principles into an accessible **community-based environment**, where emotional regulation, sensory safety, and gradual social reconnection can emerge through spatial experience and mediated interaction with reptiles.

From a design perspective, safety constitutes the primary foundational layer, encompassing biological safety, psychological safety, legal compliance, and ecological ethics. Spatial zoning would strictly separate public circulation areas, mediated interaction zones, and animal care back-of-house systems to prevent uncontrolled contact, stress, or risk. Therapeutic interaction areas would be designed for **one-to-one or small-group sessions**, allowing predictability, limited sensory stimulation, and controlled exposure—conditions particularly suited to individuals with heightened sensory sensitivity, anxiety, trauma, or autism.

Low-arousal architectural strategies form the second core design layer. These include diffuse natural lighting, non-reflective surfaces, sound-absorbing materials, stable thermal conditions, and slow spatial transitions. Such environments directly reflect attachment and right-hemispheric regulation theories by minimizing sensory overload and supporting bottom-up emotional stabilization. Transparency through glass partitions enables **visual connection without forced physical proximity**, reinforcing a sense of control and psychological safety for users.

The **reptile-assisted interaction system** functions as a mediating interface rather than a spectacle. Terrarium modules are designed as ecological micro-environments that replicate natural habitats while remaining strictly regulated in terms of temperature, humidity, UV exposure, and hygiene. Interaction occurs through **guided observation, optional tactile contact under supervision, feeding rituals, and maintenance participation**, which translate cognitive–behavioral principles into embodied experience: predictability, responsibility, attentional anchoring, and gradual emotional engagement. The absence of sudden movements, vocalization, and intrusive behavior makes reptiles uniquely compatible with low-intensity therapeutic settings.

From a cognitive–behavioral design viewpoint, the space itself operates as a **behavioral scaffold**. Circulation paths follow progressive emotional gradients from public to private, from active to calm, enabling users to regulate distance, exposure, and engagement autonomously. Reflection zones, quiet rooms, and guided breathing or grounding stations support CBT-based self-regulation, while shared observation areas gently encourage non-verbal social coexistence without immediate social demand.

Educational and ethical design layers are embedded throughout the space. Public exhibition narratives, interactive digital explanations, and workshops address animal welfare, legal frameworks, species conservation, and environmental ethics, transforming fear, stigma, and exoticism into knowledge-based understanding. This directly responds to the documented neutrality and uncertainty in public attitudes toward non-traditional companion animals. In this sense, the space operates not only as a therapeutic environment but also as a **civic platform for animal–human relational literacy**.

At the urban scale, locating such a space within the Nantes community ensures accessibility for families, schools, therapists, and social institutions. It creates a new typology that lies between clinic, educational center, and public cultural infrastructure—bridging therapy and everyday life. By integrating reptile-assisted intervention into a carefully designed public environment, this project demonstrates how emotional regulation, behavioral support, and inclusive care for vulnerable populations can be achieved through **spatial choreography, multisensory control, and ethically mediated human–animal interaction**. The design therefore becomes not a passive container for therapy, but an active agent in the co-construction of safety, attachment, and emotional recovery.

5.3 Summary

This research demonstrates that **human emotion and behavior are not governed solely by personality traits or willpower, but instead arise from a dynamic regulatory system jointly shaped by neurobiological mechanisms, attachment structures, and cognitive processing**. The right hemisphere and the autonomic nervous system primarily regulate emotional arousal, empathy, and non-verbal communication, while the cognitive system continuously modulates emotional and behavioral outputs through beliefs, schemas, and meaning-making processes. When this regulatory system is disrupted by early attachment failure, neglect, or trauma, or later distorted by maladaptive cognitive processing, individuals may develop emotional dysregulation, impulsive behaviors, anxiety disorders, autism-related symptoms, and social withdrawal. Psychological and behavioral disorders should therefore be understood not as “bad habits,” but as external manifestations of a dysbalanced neuro–attachment–cognitive regulatory system.

Within this integrated framework, Animal-Assisted Therapy (AAT) emerges as a powerful cross-species regulatory intervention. Substantial empirical evidence confirms that therapy animals can function as **alternative attachment figures**, reactivating right-hemispheric emotional regulation and parasympathetic autonomic pathways through stable, non-judgmental, and predictable interaction. This process leads to rapid reductions in anxiety, stress, and pain, alongside long-term improvements in emotional stability, behavioral control, and social functioning. These effects are reflected not only in subjective psychological outcomes but also in objective physiological indicators, including heart rate, blood pressure, cortisol, endorphins, and oxytocin. Cognitive theory further demonstrates that interaction with animals interrupts automatic negative cognitive loops and reconstructs dysfunctional schemas related to self, others, and the environment.

From a socio-historical perspective, affective bonds between humans and animals are not a modern cultural invention, but a deeply rooted evolutionary adaptation. Archaeological evidence from early sedentary societies over 12,000 years ago already demonstrates stable human–animal emotional relationships. This evolutionary continuity indicates that animal-mediated emotional regulation constitutes a long-standing biological and social mechanism. Accordingly, AAT should be understood not merely as a therapeutic technique, but as an ethical expression of **human responsibility toward life, vulnerability, coexistence, and care**. Caring for animals simultaneously becomes a process of repairing the human emotional system itself, reinforcing social responsibility and collective psychological well-being.

Within this bio–psycho–social structure, **spatial design is repositioned as an active therapeutic agent rather than a passive container**. Architectural and environmental variables—such as light, sound, scale, temperature, boundaries, and materiality—directly influence autonomic arousal, emotional safety, and attachment perception. Space thus becomes a continuous co-regulator of emotional stabilization. Through predictable layouts, controllable boundaries, and clear functional zoning, design actively participates in reconstructing core cognitive beliefs about environmental safety, self-efficacy, and emotional control.

In the specific context of reptile-assisted therapy, the low-arousal, low-invasiveness, and high predictability of reptiles aligns seamlessly with a **low-stimulation therapeutic spatial logic**, particularly suited for individuals with autism spectrum disorder, sensory integration difficulties, and anxiety disorders. The coupling of reptiles with low-contrast visual environments, stable thermal microclimates, slow spatial transitions, and transparent yet non-intrusive enclosures generates a fundamentally different therapeutic typology from dog-centered AAT environments. This combination establishes an alternative, non-verbal, low-pressure pathway for emotional reconstruction and self-regulation.

Ultimately, this research reframes animal-assisted therapy as a **highly integrated system linking neural regulation, emotional recovery, behavioral restructuring, social responsibility, and spatial design**. Within this system, design is no longer secondary but becomes a structural participant in psychological transformation. Future therapeutic spaces—particularly within urban community contexts such as Nantes—should be conceived as everyday mental health infrastructures embedded in daily life, where emotional repair, social reconnection, and interspecies coexistence unfold as continuous public experiences rather than isolated clinical events.

5.3 Strategies

This project establishes a low-arousal, low-invasiveness, and highly predictable therapeutic spatial system, in which the environment functions as a "third regulator," working in synergy with animal-assisted intervention to achieve the spatial translation of emotional stabilization, behavioral reconstruction, and secure attachment.

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